

SUMMARY

Software Development Engineer II at Amazon building cloud platforms, AI agents, data pipelines, and search systems in Python, Java, TypeScript, SQL, and AWS. Owns ambiguous projects from stakeholder requirements and design through implementation, launch, monitoring, and operations for business, science, and engineering teams.

SKILLS

- **Programming:** Python, Java, TypeScript, SQL
- **Cloud & Infrastructure:** AWS CDK, SageMaker, Bedrock, Step Functions, Glue, EMR, Redshift, Athena, Docker, CI/CD
- **Systems:** Distributed Systems, Observability, Access Control, Monitoring, Data Quality, A/B Testing
- **AI, ML & Data:** Agentic Systems, MCP, Agent Evaluation, LLM-as-Judge, MLOps, Spark, Causal Inference
- **Languages:** English, Japanese (JLPT N1), Mandarin Chinese

EXPERIENCE

- **Amazon (Japan Consumer Innovation)** Tokyo, Japan
Machine Learning Engineer July 2025 – Present
 - Architected and operate JP Retail MCP, an AI-powered business intelligence platform serving 400+ active users across 6+ departments; partnered with retail, science, and engineering stakeholders to turn metric and workflow needs into production software.
 - Built the online query path, offline ETL path, domain skill warehouse, security controls, and observability, including SQL validation with sqlglot AST parsing, team-based ACLs, structured logging, caller attribution, and CloudWatch metrics.
 - Designed an automated agent evaluation framework with deterministic graders, SQL and tool-call validation, partial-credit result checks, and LLM-as-Judge scoring; made it the release gate for new agent capabilities and regression protection.
 - Built a reusable AWS CDK data-access platform that lets 8 teams connect project-owned AWS accounts to Amazon's internal data warehouse, reducing onboarding from weeks to 1 day with standardized Redshift, Glue, EMR, IAM, and Lake Formation provisioning.
 - Delivered production ETL infrastructure for promotion benchmarking data with Step Functions orchestration, S3 event triggers, CloudWatch alarms, automated ticketing, code review gates, and developer tooling.
- **Amazon (Product Search)** Tokyo, Japan
Software Development Engineer Apr 2023 – June 2025
 - Designed and built an offline evaluation system for search refinement selection, using traffic replay, feature flags, paired statistical analysis, relevance scoring, duplication analysis, and continuous monitoring to give scientists and engineers rapid pre-A/B quality signals.
 - Contributed to the launch and worldwide expansion of a Lucene-based refinement selection backend replacing a legacy navigation system; owned pre-launch evaluation, launch readiness, change management, and validation across regional rollout phases.

EDUCATION

- **Tokyo Institute of Technology** Tokyo, Japan
M.Eng. in Information and Communication Engineering Sep 2020 – Sep 2022
- **Huazhong University of Science and Technology** Wuhan, China
B.Eng. in Electronic Information Engineering Sep 2016 – Jun 2020